Code Coverage Report

Statement, line and function coverage summary

Event Management System (Eventify)

1. Objective

This report presents measured automated test coverage for the backend and the limited frontend module that was instrumented. Coverage percentages are snapshots from the recorded coverage runs; frontend source-module coverage is not represented as whole-application UI coverage.

2. Backend overall coverage

Metric	Covered / total	Coverage
Statements	669 / 800	83.62%
Branches	409 / 491	83.29%
Functions	73 / 84	86.90%
Lines	624 / 744	83.87%

Command: from the server directory, `npm test -- --coverage`. Jest reported 21 passed suites, 1 skipped suite; 289 passed tests, 6 skipped tests; 0 failed.

3. Backend module coverage details

Module / area	Statements	Branches	Functions	Lines
Controllers (aggregate)	78.58%	81.13%	82.35%	78.35%
adminEventController	20.68%	0%	0%	20.68%
adminUserController	20.51%	0%	0%	21.62%
registrationIntegrity.js	44.73%	50.00%	55.55%	45.94%
auditLogger.js	72.72%	50.00%	100.00%	77.77%
errorHandler.js	53.33%	42.85%	100.00%	61.53%
authController	100%	—	—	100%
event browse controller	93.54%	—	—	—
event controller	94.23%	—	—	—
notification controller	100%	—	—	—
models / routes / constants / validators	100% in reported categories	—	—	—

A dash indicates that a module-specific value was not captured in the available coverage evidence; it does not mean 0%.

4. Frontend coverage

Scope	Lines	Branches	Functions	Interpretation
client/src/.../ registrationState.js	76.56%	71.43%	100%	Application source module measured
All reported frontend files combined	87.29%	75.00%	100%	Includes test file; do not present as source-only coverage

Command: from client directory, `npm test -- --experimental-test-coverage`. The test run reported 3 passed tests and 0 failures. The source module had uncovered lines 17–18, 36–37, 39–40, 42–43 and 57–63.

5. Coverage interpretation

- Backend line coverage is 83.87%; statement coverage is 83.62%; function coverage is 86.90%.

- Backend coverage is uneven across modules. Admin event and user controllers are the most significant measured gaps, with line coverage close to 21%.
- Registration integrity, audit logger and error handler contain uncovered paths that merit targeted tests.
- Frontend coverage is limited to registrationState.js and should not be interpreted as coverage of every React page/component.

6. Recommended improvements

1. Add tests for Admin list/get/update handlers and their validation, authorization and error branches.
2. Exercise registration transaction failure, replica-set error and recovery/backfill paths in registrationIntegrity.js.
3. Add audit logger tests for missing actor/event details and logger failure behavior.
4. Add error-handler tests for unhandled errors, known HTTP errors and malformed request paths.
5. Extend frontend registration-state tests to cover uncovered boundary and error branches; consider UI integration tests for components outside this measured module.

7. Conclusion

The backend reached 83.87% line coverage and 86.90% function coverage in the measured run. The frontend registration state source module reached 76.56% line and 100% function coverage. The remaining gaps are concentrated in Admin controllers and selected integrity/audit/error paths.